InternSync – Key Results

Updated Oct 29, 2026, at the end of Increment 5 (final delivery) · Author: Salam Alsaour

Every measured result of the project in one place, for the final presentation. Each number is copied from a test report or a run of a script in the repository, with its target and the way to repeat it. Numbers measured on the local Docker stack (a laptop) say so; they are repeated on the reference deployment once it is switched on (Deployment).

1. Headline Numbers

What	Result	Target
Automated tests	**202 backend tests**, **19 frontend unit tests**, **34 browser tests** on four browser engines and sizes, all passing	All pass
Code covered by the backend tests	**95%**	70%
Permission checks	**810** calls by users who must be refused, over **163** endpoints: **0** got through	0 (O1)
Matching: a relevant vacancy in the top 5	**100%** of profiles on the tuning set and **100%** on the held-out set (a random order: 80% and 36%)	80% (O2)
Skills found in CVs	**97.2%** (English 97.4%, Arabic 96.9%)	70%
Copied reports flagged	**10 of 10** on the tuning set; **7 of 10** on the held-out set: all 7 copied word for word, none of the 3 reworded	9 of 10 (O5)
Original reports wrongly flagged	**0 of 10** on both sets	At most 1 of 10
50 users at once	**95%** of requests within **0.21 s**, 0 errors in 7,785 requests	Within 2 s (NFR-PER-01)
Grade sheet of 100 students	**1.0–1.3 s** Excel, **2.4–4.6 s** PDF	Under 60 s (O6)
Similarity result with 10,000 stored reports	**0.35–0.42 s**	Within 2 minutes
Security scan (OWASP ZAP)	**0** high-risk findings; the API scan clean after two fixes	0 high (NFR-SEC-05)
Known vulnerable dependencies	**0** (pip-audit, npm audit)	0 critical
Screens in both languages	**100%**: 839 texts in Arabic and in English, right to left and left to right	100% (O7)
Running cost	**0** (free tiers only)	0 (CR-01)

2. Quality and Testing

Check	Result	Repeat with	Source
Backend tests on a real PostgreSQL	108 → 158 → 187 → 201 → **202** passing at the end of Increments 1 to 5	`pytest --cov=app` in `backend/`	Test reports 1–4, Section 1
Backend line coverage	95% at the end of every increment	same	same
Permission matrix	107 → 129 → 145 → **163** endpoints; **810** refused calls checked, none allowed	same (`tests/permissions`)	same

Check	Result	Repeat with	Source
Requirements traced	125 requirements of the SRS: 90 with an automated test named after them, 32 with a recorded result, 3 open (they need people or the hosted system), none untraced	`python docs/tools/trace.py --check`	Requirements_Status.md
Acceptance tests	Increment 1: 43 of 43 · Increment 2: 27 of 27 · Increment 3: 21 of 21 · Increment 4: 12 of 13; AT-AI-03 is met for matching and not for reworded copies (7 of 10 flagged, target 9)	same	Test reports, Section 3
Demo scenarios automated end to end	M4, M5, and M6 scenarios run through the API in the test suite	`pytest tests/test_demo_m*.py`	`backend/tests`
Browser tests	**34 passed** on Chromium, WebKit (the Safari engine), an Android phone size, and an iPhone size, including a keyboard-only task; Firefox runs in CI	`npx playwright test` in `frontend/`	Test report 4, Section 4
Frontend	TypeScript strict build clean; 19 unit tests; production build succeeds	`npx tsc -b && npx vitest run && npx vite build`	same

3. AI Accuracy

All sets are synthetic and labelled by the author (SRS C-09). The tuning sets were used to choose the settings; the **held-out sets** were written after the settings were frozen and run once, so they are the honest numbers.

Measure	Tuning set	Held-out set	Target
Skill extraction recall (20 CVs, Arabic and English)	97.2%	–	70%
Matching, relevant vacancy in the top 5	100% (9 eligible vacancies per profile; random order 79.7%)	100% (15 eligible per profile, one skill per vacancy; random order 35.7%)	80%
Matching, relevant vacancy first	100%	60% (random order 7.3%)	–
Similarity, copied reports flagged	10 of 10 (44–54% copied)	7 of 10: 7 of 7 copied word for word, 0 of 3 reworded sentence by sentence	9 of 10
Similarity, original reports flagged	0 of 10	0 of 10	At most 1

What the numbers mean:

- The similarity check finds copied text, not copied ideas. A report reworded sentence by sentence is not flagged. This is the expected limit of the word n-gram rule (Design 6.3).
- The flag is advisory: a supervisor decides, and nothing changes automatically (NFR-AI-01).

Repeat with:

```
backend/.venv/Scripts/python ai-eval/evaluate.py            # tuning sets
backend/.venv/Scripts/python ai-eval/evaluate.py --heldout  # held-out sets
```

4. Speed

Requirement	Measured	Limit	Where	Source
NFR-PER-01: ordinary requests, 50 users at once	95th percentile **0.21 s** over 7,785 requests in 5 minutes; slowest kind (placement list) 0.52 s; 0 errors	2 s	Local Docker stack	Test report 4, Section 4

Requirement	Measured	Limit	Where	Source
NFR-PER-02: ranked vacancies, 1,000 open	95th percentile **0.83 s** with the model, 0.76 s rule-based	5 s	Test database	Test report 2
NFR-PER-03: skills from a 5 MB CV	**2–6 s** (94 s before the fix found in testing)	30 s	Test database	Test report 2
NFR-PER-04: similarity with 10,000 stored reports	**0.35–0.42 s**	2 minutes	Test database	Test report 3
NFR-PER-05: grade sheet of 100 students	**1.0–1.3 s** Excel, **2.4–4.6 s** PDF	60 s each	Test database	Test report 3

5. Security and Privacy

Check	Result
OWASP ZAP baseline of the web app	0 high; medium findings only for headers that the development server does not send. The hosted app sends them (`vercel.json`), and the Content-Security-Policy was added in this increment
OWASP ZAP scan of the API (safe mode, 120 rules)	0 high; two low findings fixed (`nosniff`, resource policy), then 0 warnings
Dependency audits in CI	0 known vulnerabilities (pip-audit, npm audit)
Passwords and sign-in	Argon2id; lockout after 5 failures; 30-minute idle limit, also on the device offline
Who sees what	Every endpoint checked for every role (Section 2); the company never sees the student's evaluation of it; the grade is never on the certificate page
Privacy	Consent notice at first login; push messages carry a title only; anonymization of closed cycles after the retention period, with the statistics unchanged; offline copies deleted at sign-out

6. Languages, Devices, and Accessibility

Check	Result
Two languages	839 texts in each of Arabic and English, checked equal by a test; Arabic right to left; Arabic PDFs with joined letters
Screen widths	No horizontal scrolling at 360, 768, 1366, and 1920 px on the main pages of the student and the coordinator (browser tests)
Contrast	No text below 4.5:1 on the main pages (automated axe check); one dev-only failure fixed
Installable app and offline mode	Installs as an app; offline, pages opened earlier stay readable and drafts typed offline are saved after reconnecting (browser test on four engines)

7. Scope Delivered

Priority	Delivered
Must (Increments 1 and 2)	All
Should (Increment 3, plus the changed FR-PWA-03 in Increment 4)	All
Could (Increment 4): certificates, evaluation of the company, job-offer survey, leniency alert, anonymization	All
Product functions F1 to F19 of the Proposal	19 of 19

8. Still to Measure

What	When
User acceptance testing, at least 8 participants, 80% of scenarios without help (O8)	Postponed by the owner on Oct 29; the plan is ready to run at any time (plan)
Install, push, and airplane mode on real Android and iPhone devices	In the UAT sessions
NFR-PER-01 and the ZAP scan on the reference deployment	After the switch-on (Deployment)

Revision History

Date	Change
Oct 28, 2026	First version, at the end of Increment 4
Oct 29, 2026	Increment 5: 202 backend tests; the UAT postponed; requirements status in Section 2